

Title : MongoDB Aggregation & Indexing

Objective :-

To acquire the skills to use a Powerful, Flexible & scalable general Purpose database to handle Bigdata.

Outcome :- Implement No SQL queries using MongoDB

Software :- MongoDB

Theory :-

INDEXING

Indexes support the efficient resolution of queries without indexes, MongoDB must scan every document of a Collection to select those documents that match the query statement.

This scan is highly inefficient & requires the MongoDB to process a large volume of data.

Indexes are special data structures that store a small portion of data set in an easy to traverse form."

The index stores the values of a specific field or set of fields, ordered by the value of the field as specified in index.

The ensure Index() Method :-

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To create an index you need to use ensure Index() method of mongoddb.

Syntax :-

Basic syntax of ensure Index() method is as follows()
`> db.COLLECTION_NAME.ensureIndex({KEY:1})`

Here key is the name of field on which you want to create index 0 & 1 is for ascending order. To create index in descending order you need to use -1.

eg :-

`> db.mysql.ensureIndex({ "title" : 1 })`

In ensure Index() method you can pass multiple fields, to create index on multiple fields.

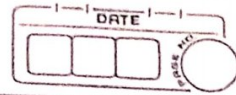
In ensure Index() method you can pass multiple fields, to create index on multiple field.

`> db.mycol.ensureIndex({ "title" : 1 "description" : -1 })`

Drop Index :

`> db.mycol.dropIndex({ "title" : 1, "description" : -1 })`

AGGREGATION :



- Aggregation operations process data records & return computed results.
- Operation on the grouped data to return a single result. In Sql Count (*) & with group by is an equivalent of MongoDB aggregation.
- The aggregate () method for the aggregation in mongodb you should use aggregate () method.

Syntax :

Basic Syntax of aggregate () method is as follows

```
> db.mycol.aggregate ( [ { $group : { _id : "$ by user",  
num_tutorial : { $sum : 1 } } } ] )  
{ "result" : [ { "_id" : "tutorial Spoint", "num-  
tutorial" : 2 } ;  
{ "_id" : "tutorials point", "num_tutorial" : 3 }  
, "ok" : 1 }
```

Conclusion :-

We have implemented aggregation & indexing using mongodb.

FOR EDUCATIONAL USE